

SGU X ROCKET RIDE HACKATHON

ROUND 1 PRESENTATION

COLLEGE NAME : IMSEC / BBDIT

Team name : CODING DIVAS

Team leader : Aadya Dixit

Team Member: Khushbu

Problem Statement : Deepfake & Digital Media Authenticity Verification

PROBLEM STATEMENT

DEEPFAKE & DIGITAL MEDIA AUTHENTICITY VERIFICATION PLATFORM

- ♦ Generative AI has made it difficult to distinguish real from AI-generated images, videos, and audio.
- ♦ Deepfakes are increasingly used for misinformation, fraud, identity theft, and reputation damage.
- ♦ Existing verification methods are slow, costly, and not easily accessible.
- ♦ There is a need for an AI-powered, scalable, and automated platform to verify digital media authenticity with a confidence score before or after sharing.

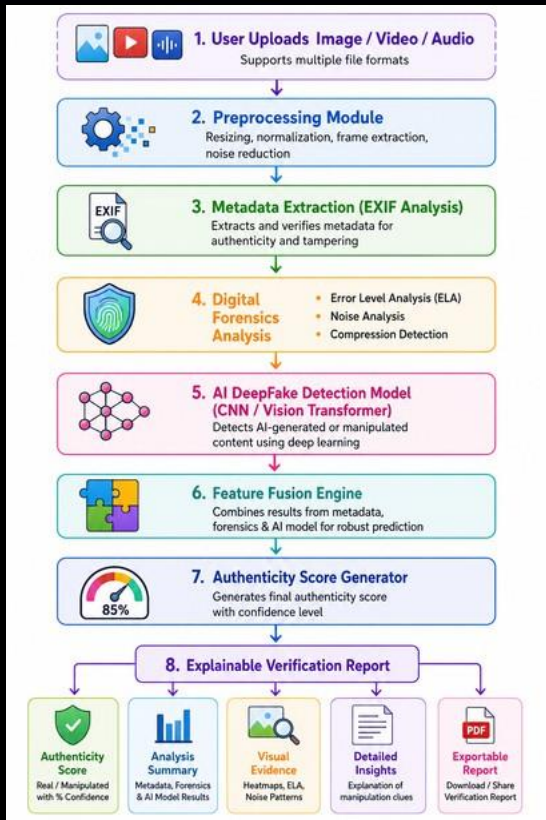


PROPOSED SOLUTION

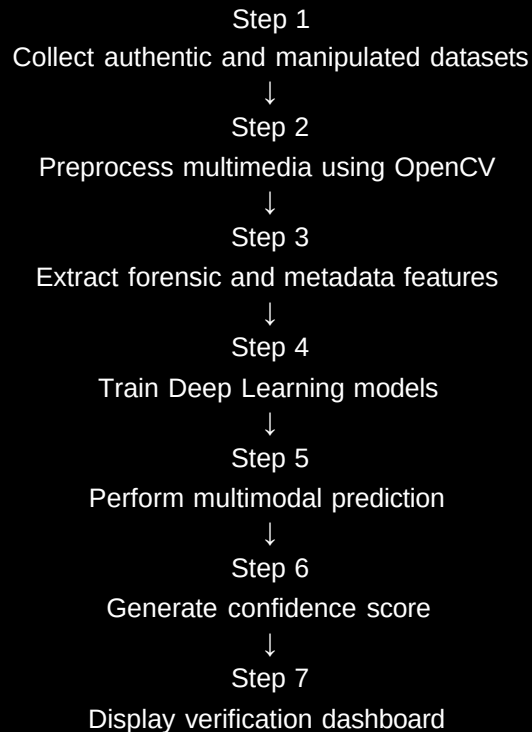
- 1. Our solution is AI-powered multimedia verification platform
- 2. Detects AI-generated images, videos, and audio
- 3. Performs metadata and digital forensic analysis
- 4. Identifies editing artifacts and visual inconsistencies
- 5. Generates an AI authenticity confidence score
- 6. Provides Explainable AI (XAI) verification reports
- 7. Supports pre-sharing and post-sharing content verification



TECHNICAL APPROACH



Process of implementation



FEASIBILITY AND VIABILITY

FEASIBILITY ANALYSIS

AI-Based DeepFake Detection System

TECHNICAL FEASIBILITY

- ✓ Availability of large public DeepFake datasets
- ✓ Mature AI frameworks (TensorFlow, PyTorch)
- ✓ Open-source Computer Vision libraries (OpenCV, Dlib, FFmpeg)
- ✓ Cloud deployment support (AWS, Azure, GCP)



ECONOMIC FEASIBILITY

- ✓ Low development cost
- ✓ Open-source software stack
- ✓ Scalable cloud infrastructure
- ✓ Suitable for startups and enterprises



OPERATIONAL FEASIBILITY

Applicable in:

- | | |
|--------------------------|--------------------------|
| Social Media Platforms | Digital Journalism |
| News Agencies | Banking & Finance |
| Government Organizations | Educational Institutions |
| Cybercrime Investigation | |



This system is technically sound, cost-effective, and practically applicable across multiple industries to combat DeepFake threats.

Viability

- ◆ Addresses the growing need for digital media authentication and deepfake detection.
- ◆ Helps prevent misinformation, fraud, and identity misuse.
- ◆ Applicable in cybersecurity, journalism, law enforcement, and enterprises.
- ◆ Provides a scalable solution for secure and trustworthy digital communication.

METHODOLOGY

- ✓ AI-generated Image Detection
- ✓ DeepFake Video Detection
- ✓ Synthetic Voice Detection
- ✓ Metadata Verification
- ✓ Error Level Analysis (ELA)
- ✓ Image Noise Analysis
- ✓ Confidence Score Generation
- ✓ Explainable AI Report
- ✓ User-friendly Dashboard
- ✓ Cloud-based REST API
- ✓ Secure File Upload
- ✓ Real-time Verification

TECH STACK

 Category	 Technology / Tools
 Programming	 Python
 Machine Learning	  TensorFlow, PyTorch
 Computer Vision	 OpenCV
 Deep Learning	 CNN, Vision Transformer (ViT), EfficientNet
 Audio Processing	 Librosa
 Data Analysis	  NumPy, Pandas
 Backend	  Flask / FastAPI
 Frontend	 React.js /  Streamlit
 Database	 MongoDB /  PostgreSQL
 Cloud	 /  /  /  GCP
 Deployment	 Docker
 Version Control	 Git, GitHub 

IMPACT AND BENEFITS

Impact

- ◆ Reduce misinformation
- ◆ Increase digital trust
- ◆ Strengthen cybersecurity
- ◆ Assist journalists and investigators
- ◆ Improve media authenticity verification
- ◆ Support governments and organizations in combating AI-generated fraud

Benefits

- ◆ Provides fast and accurate media authenticity verification.
- ◆ Protects users and organizations from identity misuse.
- ◆ Enables reliable and secure digital communication.

SOURCES

The idea is inspired by the growing misuse of Generative AI technologies and the increasing need for reliable digital media verification

Research Sources

- Google DeepMind research on AI-generated media
- OpenAI research on Generative AI
- Meta AI DeepFake Detection Challenge
- FaceForensics++
- DeepFake Detection Challenge (DFDC)
- Kaggle DeepFake datasets
- IEEE Research Papers
- ACM Digital Library
- CVPR & ICCV Computer Vision publications
- Cybersecurity reports from NIST and CISA

LINKS:

□□ Project Documentation Link:

<https://drive.google.com/file/d/1nSnvIPRMiSRlkxZHVSOFoifYg4Jv3yrp/view?usp=drivesdk>

□□ GitHub Repository Link:

<https://github.com/aadyadixit17-code/DeepFake-Digital-Media-Authenticity-Verification>

□□ Working Demo Link:

https://drive.google.com/file/d/1SnKgiu5cUtlZSPiOw0iM5paNZ7tB_5Pi/view?usp=drivesdk

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